Project Proposal: Multimodal Presurgical Brain Tumor Classification

STAT3612 Statistical Machine Learning — Spring 2026

Group Members: [Member 1], [Member 2], [Member 3], [Member 4], [Member 5]

1. Project Overview

This project aims to develop statistical machine learning models for presurgical brain tumor classification into five categories (brain metastasis, choroid plexus tumour, glioma, meningioma, and pineal/sellar region tumours) using multimodal data. Our approach follows a three-stage pipeline: (1) single-modality baselines, (2) multimodal feature fusion, and (3) ensemble optimization with comprehensive model analysis. We plan to leverage all three available data modalities — MRI-derived imaging features (radiomics and deep features), clinical/demographic information, and radiology reports — to maximize classification performance.

2. Proposed Methods

Stage 1: Single-Modality Baselines

We will first build independent classifiers for each data modality to understand their individual contributions:

- **Radiomic features:** Apply dimensionality reduction (PCA, feature selection via mutual information) followed by XGBoost, Random Forest, and SVM classifiers.
- **Deep image features (ResNet):** Train classifiers (Logistic Regression, MLP, XGBoost) on the pre-extracted deep features. Explore fine-tuning a lightweight head if beneficial.
- **Clinical & text data:** Encode demographics (age, gender) and extracted clinical variables. For raw radiology reports, extract features using TF-IDF and keyword-based approaches; explore lightweight embeddings if time permits.

Stage 2: Multimodal Fusion

- **Early Fusion:** Concatenate features from all modalities after normalization and dimensionality reduction, then train unified classifiers (XGBoost, SVM, MLP).
- **Late Fusion:** Combine predictions from single-modality models via weighted voting or a meta-learner (stacking with Logistic Regression or XGBoost as the second-level model).

Stage 3: Ensemble Optimization

Build a final stacking ensemble that integrates the best-performing models from both fusion strategies. Address class imbalance through SMOTE, class-weighted loss functions, and stratified cross-validation. Perform hyperparameter tuning via Bayesian optimization (Optuna).

3. Model Analysis Plan

To ensure comprehensive evaluation (targeting the 40% analysis component), we plan the following analyses:

- **Model comparison:** Compare all candidate models (RF, XGBoost, SVM, MLP, Stacking) using F1, AUC, precision, recall, and accuracy on the validation set.
- **Modality ablation study:** Systematically evaluate performance with different modality combinations to quantify each modality's contribution.
- **Interpretability analysis:** Use SHAP values and feature importance rankings to identify the most discriminative features across tumor types.
- **Class-wise performance:** Report per-class F1, precision, and recall; analyze confusion patterns between tumor types.
- **Class imbalance analysis:** Compare the effects of different resampling strategies (SMOTE, random oversampling, class weights) on minority class performance.

4. Timeline

Phase	Period	Key Tasks

Data Exploration & Baselines	Apr 6 – Apr 10	EDA, preprocessing, single-modality baseline models, first Kaggle submission
Multimodal Fusion & Optimization	Apr 11 – Apr 17	Feature fusion, ensemble methods, hyperparameter tuning, class imbalance
Final Model & Analysis	Apr 18 – Apr 20	Finalize best model, run ablation study, SHAP analysis, per-class evaluation
Report & Slides	Apr 21 – Apr 23	Write report, prepare slides, clean Jupyter Notebook, rehearse
Presentation	Apr 24	Oral presentation, final Kaggle submission

5. Work Distribution

Member	Primary Responsibility
Member 1	Radiomic features processing, traditional ML baselines (RF, SVM)
Member 2	Deep image features, MLP/neural network experiments
Member 3	Text/NLP feature extraction from radiology reports
Member 4	Multimodal fusion, ensemble methods, hyperparameter tuning
Member 5	Model analysis (SHAP, ablation), report writing, slides preparation

Note: All members will collaborate on cross-cutting tasks such as Kaggle submissions, code review, and presentation rehearsal.

6. Resources

[1] He K, Zhang X, Ren S, et al. Deep Residual Learning for Image Recognition. CVPR, 2016.
[2] Chen T, Guestrin C. XGBoost: A Scalable Tree Boosting System. KDD, 2016.
[3] Lundberg S, Lee S. A Unified Approach to Interpreting Model Predictions (SHAP). NeurIPS, 2017.
[4] PyRadiomics Documentation: https://pyradiomics.readthedocs.io/en/latest/
[5] Scikit-learn, XGBoost, Optuna, NLTK — Python ML and NLP libraries.
[6] Google Colab — GPU resources for deep learning experiments.